DEPARTMENT OF ELECTRONICS &COMMUNICATION ENGINEERING UNIVERSITY COLLEGE OF ENGINEERING (A)

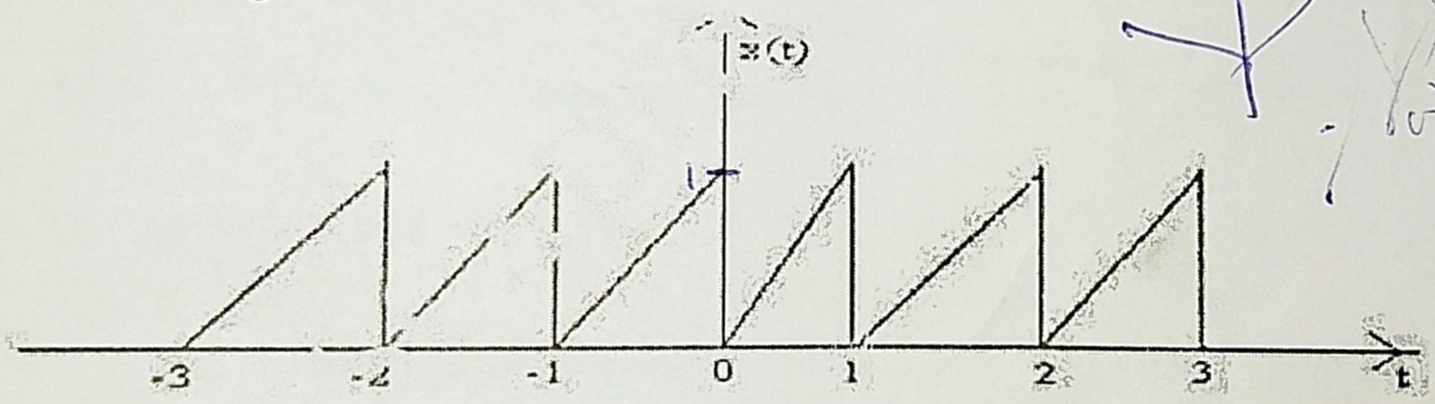
Osmania University, Hyderabad-07

CLASS TEST-I

SUBJECT :SIGNAL AND SYSTEMS

Class: BE (CSE)
Semester: IV
Academic Year:2026-27

Max Marks: 30
Duration: 90 Min

		Marks	BT	CO
	PART A Answer All Questions (10 M)			
1. a)	Sketch the following signals i)u(-t+2) ii) -4r(t)	2	1	1
b)	Check whether the following systems are periodic or not? If periodic determine the fundamental period i)u(t)- 2u(t-5) ii)sin12Πt	2	2	1
c)	Write Dirichlet's condition of Fourier Series	2	3	2
d	Write Linearity property of Fourier Series	2	4	2
e	Define and discuss various elementary continuous time signals. Indicate them graphically	2	1	1
	PART B Answer Any Two Questions (20 M)			
2. a)	Find the Trigonometric Fourier series for the saw tooth waveform shown in figure 	10	2	2
3. a)	Check whether the following systems are i)static or Dynamic ii)Linear or Non linear iii)Causal or Non causal iv)Time invariant or time variant 1. $y(n) = x(n)\, x(n-2)$	8	3	1
b	Prove that the power of the energy signal is zero over infinite time	2		1
4. a)	Prove that the Fourier series expansion of an odd periodic function contains only sine terms.	5	4	2
b)	State and prove the i) convolution theorem property ii) Time differentiation property of Fourier series	5	5	1

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